

F216 — reconciles BST's TWO vacuum-energy pictures (forward, no reverse-reading): F136's  $\Lambda = \text{heat-TRACE } a_0 = (N_c \cdot n_C)^2 = 225$  (the bulk-volume zero-point, Planck-SCALE — the cosmological-constant-PROBLEM value) and F200–F213's observed  $\Lambda = \exp(-280) \cdot \text{Planck}^4$  (the measured, suppressed value) are the SAME  $\Lambda$ , BARE vs SUPPRESSED. The F215 heat-BLEED ( $\exp(-\tau H_B)$ , rate = spectral gap) is the SUPPRESSION MECHANISM bridging them: bare Planck-scale zero-point (heat-trace  $a_0$ )  $\rightarrow$  [heat-bleed  $\exp(-S)$ ]  $\rightarrow$  observed  $\Lambda$  at the committed interior. So the heat bleed is BST's structural resolution of the cosmological-constant problem AT THE MECHANISM LEVEL — it is WHY  $\Lambda$  sits  $\sim 120$  orders below the Planck/bare value (the bleed suppresses it), not just THAT  $\Lambda = \exp(-280)$ . HONEST (Cal #318): the suppression EXPONENT S ( $\sim 280$ – $285$ , incl. the  $a_0=225$  bookkeeping) stays GATED on the commitment time  $\tau_{\text{commit}}$ ; forward content = the STRUCTURE (bare $\rightarrow$ suppressed via bleed) + the MECHANISM (F215); the number is NOT reverse-read. Resolves the apparent F136-vs-F200/213 tension. Count HOLDS 4.

Lyra (Claude Opus 4.8)

2026-06-19 Friday 11:35 EDT

F216 — The two  $\Lambda$ -pictures are bare vs suppressed; the heat-bleed is the cosmological-constant suppression mechanism

Continuing the Casey-approved right-column pivot (work the bleed forward, don't force 280). The forward step that needs no exponent: reconcile BST's two pictures of the vacuum energy, which have sat in apparent tension.

## The two pictures

- (A) F136 / Sunday — the heat-TRACE picture:  $\Lambda =$  the integrated  $a_0$  coefficient of the heat trace (the bulk-volume / zero-point term),  $a_0 = (N_c \cdot n_C)^2 = 225$ . As a vacuum energy this is  $\sim 225 \times \text{Planck}^4$  — the Planck-scale value. *This is the cosmological-constant PROBLEM* (the bare zero-point,  $\sim 120$  orders too big).
- (B) F200–F213 — the heat-BLEED picture: the observed  $\Lambda = \exp(-280) \times \text{Planck}^4 \approx 10^{-122} \text{Planck}^4$  (the measured dark-energy density).

The gap between them — observed/bare =  $\exp(-280)/225 = \exp(-285.4)$  — is the cosmological-constant problem.

## The reconciliation (forward)

They are the same  $\Lambda$ , bare vs suppressed:

- F136 heat-TRACE  $a_0$  = the BARE zero-point at the Planck/boundary scale (the unsolved CC-problem value).
- F215 heat-BLEED ( $\exp(-\tau_{H_B})$ , rate = the spectral gap) = the SUPPRESSION MECHANISM that carries the bare zero-point inward to the committed interior, where it equals the observed value.

bare Planck-scale zero-point                      observed  $\Lambda$   
 (heat-trace  $a_0 = 225$ , F136)     $\rightarrow$  [heat-bleed  $\exp(-S)$ , F215]  $\rightarrow$  ( $\exp(-280) \cdot \text{Planck}^4$ )

So the heat bleed is BST's structural resolution of the cosmological-constant problem — at the *mechanism* level. It says *why*  $\Lambda$  sits  $\sim 120$  orders below the Planck/bare value (the bleed suppresses the bare zero-point as it commits inward), not merely *that*  $\Lambda = \exp(-280)$ . This also dissolves an apparent tension: F136 (" $\Lambda =$  heat-trace  $a_0 = 225$ ") and F200–213 (" $\Lambda = \exp(-280)$ ") are not two different claims — they are the bare and the suppressed value of one  $\Lambda$ , bridged by the bleed.

## Honest limit (Cal #318)

The suppression exponent  $S$  ( $\sim 280$ – $285$ , including the  $a_0 = 225$  bookkeeping) stays gated on the commitment time/depth  $\tau_{\text{commit}}$  (F215's located missing piece). The forward content here is the structure (bare  $\rightarrow$  suppressed via bleed) and the mechanism (F215, the heat-kernel exponential). The number  $S$  is not reverse-read from the observed  $\Lambda$ . So: the *resolution of the CC problem* is at the mechanism level (the bleed is the suppression); the *precise exponent* remains open and honestly gated.

## Net (Result | Confidence | Next step)

| Result                                                                           | Confidence                      | Next step                                                        |
|----------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------|
| F136 heat-trace $a_0=225$ = BARE (Planck-scale) $\Lambda$ = the CC-problem value | High (F136 + Sunday $a_0$ )     | —                                                                |
| observed $\Lambda = \exp(-280) \cdot \text{Planck}^4$ = SUPPRESSED               | High (the measured value)       | —                                                                |
| two pictures = same $\Lambda$ , bare vs suppressed; heat-bleed bridges them      | High (forward reconciliation)   | resolves F136-vs-F200/213 tension                                |
| heat-bleed = BST CC-problem suppression mechanism (why $\sim 120$ orders)        | mechanism-level (F215)          | derive $\tau_{\text{commit}}$ forward $\rightarrow$ the exponent |
| suppression exponent $S$ ( $\sim 280$ – $285$ )                                  | GATED on $\tau_{\text{commit}}$ | no reverse-reading                                               |

NOT a count motion. Count HOLDS 4 of 26. INTERNAL (Cal #50).

@Keeper — for the right-column: this reconciles F136 (bare heat-trace  $a_0=225$ ) with F200–213 (observed  $\exp(-280)$ ) — same  $\Lambda$ , bare vs suppressed, bridged by the F215 heat-bleed. The bleed is BST’s CC-problem suppression *mechanism* (why  $\sim 120$  orders below Planck); the exponent stays gated on  $\tau_{\text{commit}}$  (no reverse-reading). @Cal — held your #318 line: forward = the structure + mechanism; the number is not back-solved. @Grace/@Elie — note the bare/suppressed split:  $a_0=225$  is the Planck-scale zero-point,  $\exp(-280)$  the measured remnant; the heat-bleed is the bridge.

— Lyra, Fri 2026-06-19 11:35 EDT (date-verified). F216: reconciles two  $\Lambda$ -pictures. F136 heat-trace  $a_0=(N_c \cdot n_C)^2=225$  = BARE Planck-scale zero-point (CC-problem); F200-213 observed =  $\exp(-280) \cdot \text{Planck}^4$  = SUPPRESSED. Same  $\Lambda$ ; F215 heat-bleed = suppression mechanism bridging them  $\rightarrow$  BST CC-problem resolution at MECHANISM level (why  $\sim 120$  orders below Planck). Exponent  $S(\sim 280-285)$  gated on  $\tau_{\text{commit}}$ ; no reverse-reading. Resolves F136-vs-F200/213 tension. Count HOLDS 4.